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The Ex-Dividend Day Price Drop Anomaly

Evidence From 19,319 S&P 500 Ex-Dividend Events (2010–2025)

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References 19 References

ABSTRACT

This study asks whether stock prices fall by the full dividend on the ex-dividend date, as frictionless pricing predicts. It reruns the analysis on real daily prices from Yahoo Finance for every current S&P 500 company that paid a dividend between 2010 and 2025, counting each company only from the date it joined the index: 19,319 ex-dividend events from 399 firms in 11 sectors. Before any adjustment, shares open lower by 0.777 times the dividend on average. Because the market itself rose slightly on the typical ex-date, the raw figure understates the drop, so the study adds back each stock's expected overnight move, using its estimated beta and the S&P 500 ETF's opening return. The market-adjusted drop averages

0.866 times the dividend (95% CI 0.831 to 0.901, with standard errors clustered by date and firm), a shortfall of about 13% that is statistically significant (t = −7.48) and holds up under the robustness checks in Section 4.2. The October 2026 revision of this paper had estimated a corrected average of about 1.0 from the original paper's own figures; the real data show that estimate was wrong, because it used the market's full-day return rather than its opening move. The shortfall is largest for low-yield stocks (0.798) and smallest for high-yield stocks (0.930), does not differ significantly across sectors or years, and is unrelated to share price. At the stock level, 32 of 343 stocks show a significant shortfall after controlling the false discovery rate, led by AT&T, Cardinal Health, Boeing, and Ventas, while none shows a significantly larger-than-full drop. A dividend capture trade (buy at the close, sell at the ex-day open) earned 9.03 basis points per event before trading costs and taxes.

KEYWORDS: EX-DIVIDEND DAY, PRICE DROP RATIO, TAX CLIENTELE EFFECT, DIVIDEND CAPTURE, MARKET MICROSTRUCTURE, CLUSTERED INFERENCE

SECTION 1

1. Introduction

In a frictionless capital market with no taxes and no transaction costs, a stock's price should decline by exactly the amount of the dividend on the ex-dividend date. This prediction follows directly from the absence of arbitrage: if the stock dropped by less than the dividend, an investor could buy shares the day before, capture the dividend, and sell at a profit; if it dropped by more, an investor could short before and cover after. The Miller & Modigliani (1961) dividend irrelevance proposition provides the theoretical foundation: in perfect markets, dividend policy is irrelevant to firm value, and the ex-day price adjustment should be dollar-for-dollar.

Yet real markets are not frictionless. At least three classes of friction may cause the observed price drop to deviate from the dividend amount:

1. Differential taxation. In most jurisdictions, dividends and capital gains are taxed at different rates. If the marginal investor faces a higher tax rate on dividends than on capital gains, the equilibrium ex-day price drop will be less than the full dividend, reflecting the after-tax indifference condition. Elton & Gruber (1970) formalized this insight and used the observed drop ratio to infer the marginal tax rate of stockholders.
2. Transaction costs and market microstructure. Bid-ask spreads, tick sizes, and order-flow dynamics can distort the observed price change. Frank & Jagannathan (1998) showed that even in Hong Kong, where dividends were tax-free, stock prices dropped by less than the dividend, implicating microstructure effects rather than taxes alone. Graham, Michaely, & Roberts (2003), however, found that as U.S. tick sizes shrank from eighths to sixteenths and then to decimals in 2001, ex-day abnormal returns rose and the median drop ratio fell, which they judged inconsistent with microstructure explanations.
3. Heterogeneous investors and dividend capture. Institutional investors, short-term traders, and tax-exempt or tax-deferred investors (pension funds and retirement accounts) may have different valuations of dividends versus capital gains. Trading around ex-dates ("dividend capture" strategies) introduces additional price dynamics (Lakonishok & Vermaelen, 1986; Michaely & Vila, 1995).

This paper contributes an updated empirical analysis using 16 years of modern market data (2010–2025), spanning the post-decimalization era and encompassing diverse market conditions including the COVID-19 crash, the 2022 bear market, and the AI-driven bull market. It uses a point-in-time S&P 500 sample, a market adjustment timed to the opening price, standard errors clustered by date and firm, stock-level tests with false discovery rate control, and a direct test of dividend capture.

SECTION II

2. Literature Review

The study of ex-dividend day price behavior dates to Campbell & Beranek (1955), who first documented that stocks tend to drop by less than the dividend amount. The seminal contribution of Elton & Gruber (1970) formalized the tax-clientele explanation: if the marginal investor's tax rate on dividends exceeds that on capital gains, the equilibrium price drop satisfies:

$$\Delta P / D = (1 - \tau_d) / (1 - \tau_g)$$

where τ_d and τ_g are the tax rates on dividend income and capital gains, respectively. They used observed drop ratios to back out implied marginal tax rates and found evidence consistent with the clientele effect: high-yield stocks (held by lower-tax-bracket investors) exhibited drop ratios closer to 1.0.

Black & Scholes (1974) approached the question from the return side. They concluded that, using the best available empirical methods, it is not possible to show that expected returns on high-yield stocks differ from those on low-yield stocks, either before or after taxes, and therefore that the effect of a change in dividend policy on a firm's stock price cannot be determined from the data.

Kalay (1982) challenged the pure tax interpretation, arguing that transaction costs create a band of no-arbitrage around the theoretical drop, making it impossible to precisely identify the marginal investor's tax rate from price data alone.

Lakonishok & Vermaelen (1986) documented abnormal trading volume around ex-dividend dates, consistent with tax-motivated trading and dividend capture strategies. They found that short-term traders systematically trade around ex-dates to exploit differential tax treatment.

Boyd & Jagannathan (1994) proposed a dynamic model where the drop ratio depends on the relative proportions of long-term investors and short-term dividend capturers in the market.

Bali & Hite (1998) argued that discrete tick sizes (1/8ths in the pre-decimalization era) mechanically prevented prices from adjusting by the exact dividend amount, since dividends rarely fell on tick boundaries. They predicted, and found, that the bias was related to the dividend's position within the tick.

Frank & Jagannathan (1998) provided a critical natural experiment: Hong Kong, where dividends were not taxed. Even there, stock prices dropped by only about 50% of the dividend amount. This evidence strongly supports non-tax explanations, particularly market microstructure effects. They proposed a bid-ask explanation: because market makers have a cost advantage in collecting dividends, they tend to buy cum-dividend at the bid and sell after the ex-date at the ask, so cum-day trades cluster at the bid and ex-day trades at the ask, which mechanically reduces the observed drop.

Graham, Michaely, & Roberts (2003) exploited the NYSE's move from eighths to sixteenths and then to decimal pricing, completed in January 2001. The median drop ratio fell from 0.893 in the eighths era to 0.833 in the sixteenths era and 0.750 in the decimal era, and ex-day abnormal returns rose, the opposite of what tick-size explanations predict. They concluded that their results are inconsistent with microstructure explanations and consistent with the tax explanation, which attributes the shortfall to the preferential taxation of capital gains relative to dividend income.

Elton, Gruber, & Blake (2005) compared taxable and non-taxable closed-end funds, finding that the tax status of investors significantly affects the ex-day price drop, providing clean support for the tax-clientele hypothesis.

Cloyd, Li, & Weaver (2006) examined the interaction of tick sizes and tax effects, finding that both independently contribute to the anomaly.

Zhang, Farrell, & Brown (2008) studied the 2003 dividend tax cut (which reduced the top rate on qualified dividends to 15%) and found that the drop ratio increased after the cut, consistent with tax-based explanations.

Rantapuska (2008), using trading data on all investors in the Finnish stock market, found that investors who prefer dividend income buy cum-dividend and sell ex-dividend while investors with the opposite preference do the reverse, and that transaction costs and dividend yield jointly determine whether any short-term ex-day trading occurs.

SECTION III

3. Data and Methodology

3.1 Sample Construction

The sample starts from the companies in the S&P 500 as of November 5, 2026, with the date each was added to the index, taken from the public constituent list maintained in the datasets/s-and-p-500-companies repository on GitHub (sourced from Wikipedia). Companies with two share classes in the index (Alphabet, Fox, and News Corp) contribute one class each, leaving 500 companies, of which 408 paid at least one dividend with an ex-date between January 1, 2010 and December 31, 2025. To keep the sample point-in-time on the way in, an event counts only if its ex-date falls on or after the company's addition to the index; this removes 3,256 events from years before a company joined. Special dividends, defined as a payment more than 2.5 times the median of the company's previous four dividends, are excluded (177 events), since they are typically announced and priced differently from regular dividends. The final sample has 19,319 ex-dividend events from 399 firms, rising from 845 events in 2010 to 1,532 in 2025 as more of today's members join the index and more companies pay dividends. The full ticker list is given in the Data Appendix.

The original version of this paper used 80 hand-picked stocks, 8 per sector, that it never listed, so that sample cannot be rebuilt; the point-in-time S&P 500 sample replaces it. One form of survivorship remains: companies that left the index before November 2026 through acquisition, bankruptcy, or removal are absent, because their price histories are often no longer served and a reused ticker cannot be matched safely to its former company. Section 5.4 discusses what this may do to the estimates.

All prices come from the Yahoo Finance chart service (daily bars, retrieved November 10, 2026), the same source the original paper used through the yfinance library. For each stock we use the daily open and close, which Yahoo adjusts for later stock splits but not for dividends, so the ex-day drop is preserved; the dividend-adjusted close, used only to estimate betas; and the full dividend and split history. Yahoo reports dividend amounts on the same split-adjusted basis as its prices, which we confirmed on known events; for example, Coca-Cola's $0.51 dividend that went ex on December 1, 2025 met a prior close of $73.12 and an ex-day open of $72.60, a drop of $0.52. The SPDR S&P 500 ETF (SPY) serves as the market proxy.

3.2 Variable Construction

For each ex-dividend event, with t the ex-date and t−1 the prior trading day:

$$DR_{open} = (Close_{t-1} - Open_t) / D \qquad DR_{close} = (Close_{t-1} - Close_t) / D$$

The raw close-to-open ratio mixes the dividend with whatever the market did overnight. Our main measure adds back the stock's expected overnight move:

$$DR_{adj} = DR_{open} + \beta \times R_{m,open} \times Close_{t-1} / D$$

where $R_{m,open}$ is SPY's return from the prior close to the ex-day open (adding back SPY's own dividend when the two ex-dates coincide), and β is the stock's beta from an ordinary least squares regression of its daily total returns on SPY's over the 252 trading days ending the day before the ex-date (at least 126 days required; the 32 events without enough history use $\beta = 1$). A rising market lifts the ex-day open and shrinks the measured drop, so the market term is added. The original paper subtracted it and used SPY's full-day return; Section 4.3 reports both of those versions for comparison.

Dividend yield is the trailing twelve-month yield: the sum of regular dividends with ex-dates in the 365 days through the event, divided by the prior close. This replaces the original paper's quarterly annualization, which understated the yield of monthly payers. Nominal share price, used in the price-level test, is the split-adjusted close multiplied back by every split that occurred after the event, which recovers the price that actually traded at the time.

3.3 Filtering

As in the original paper, ratios outside [−20, +20] are treated as data errors or news-driven outliers. Because the trim is applied to the outcome itself, we report how many events it removes for each measure (101 for the main measure, 190 for the raw close-to-open ratio) and show the untrimmed results in Section 4.2. Yahoo records the ex-day open as exactly equal to the prior close for 218 events, a share higher than chance would suggest; Section 4.2 also shows results without them.

3.4 Statistical Inference

Events on the same date share the same market shock, and events from the same firm share firm characteristics, so treating them as independent overstates precision (Petersen, 2009). All standard errors, confidence intervals, and t-statistics for means and regressions are therefore clustered two ways, by ex-date and by firm (Cameron, Gelbach, & Miller, 2011). We also report a firm-cluster bootstrap interval (2,000 resamples of whole firms), the Wilcoxon signed-rank test against 1.0, and, for comparison with the original paper, the conventional t-statistic that treats events as independent.

Differences across yield terciles, sectors, and years are tested with Wald tests on the same clustered covariance. Stock-level tests use each stock's own events, and the resulting p-values are adjusted for the number of stocks tested with the Benjamini and Hochberg (1995) false discovery rate procedure. To separate real stock-to-stock differences from noise, we also estimate the spread of true stock-level ratios with an empirical Bayes model (Efron & Morris, 1975).

SECTION IV

4. Results

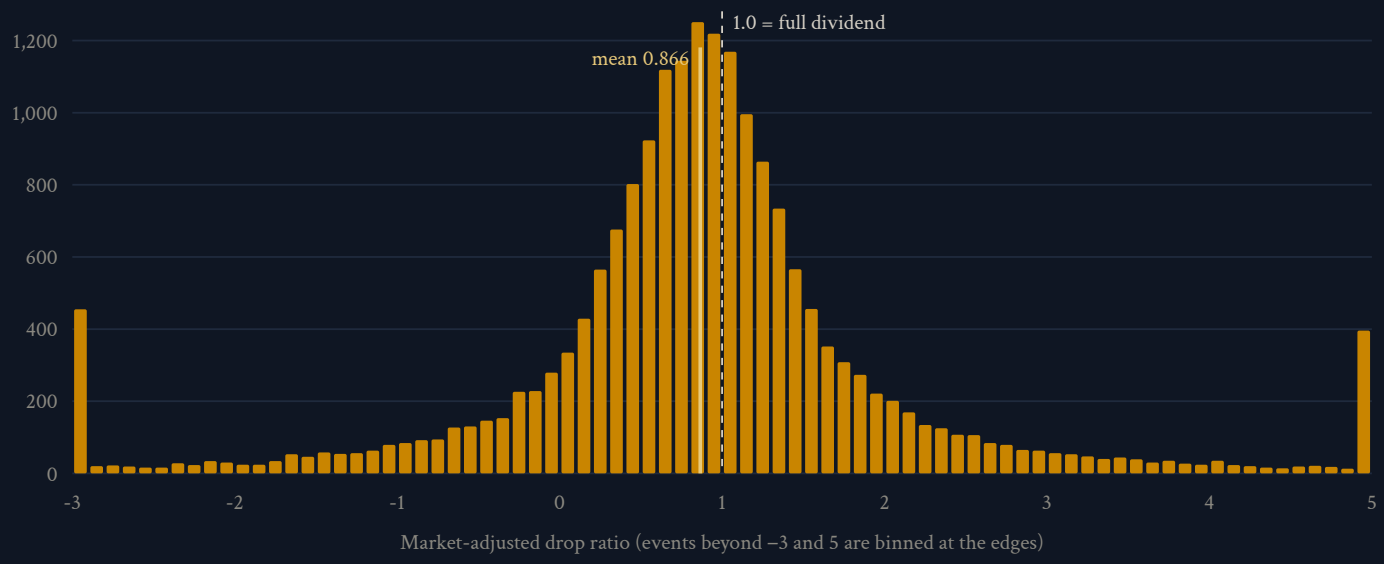
4.1 Descriptive Statistics

Table 1: Summary Statistics for Ex-Dividend Drop Ratios

Statistic	Raw close-to-open	Raw close-to-close	Market-adjusted (main)
N (after ±20 trim)	19,129	18,842	19,218
Events trimmed	190	477	101
Mean	0.777	0.749	**0.866**
Median	0.819	0.810	0.873
Std. dev.	2.601	4.044	1.950
25th percentile	0.116	−0.709	0.411
75th percentile	1.450	2.282	1.328
Skewness	−0.25	−0.16	−0.28
Kurtosis	17.8	8.0	28.1

Before any adjustment, shares open lower by an average of 0.777 times the dividend (median 0.819), and close the ex-day lower by 0.749 times the dividend. Individual events are extremely noisy: the dividend averages about half a percent of the share price, while an ordinary day's market move is of similar size, so single ratios range widely and the distribution has very heavy tails (kurtosis 28.1 for the main measure). The market adjustment pulls the mean up to 0.866 and, by removing the shared market move, cuts the standard deviation from 2.60 to 1.95.

FIGURE 1. DISTRIBUTION OF MARKET-ADJUSTED DROP RATIOS



All 19,218 events after the ±20 trim, in bins of 0.1. Values below −3 or above 5 are gathered in the end bins. The gold line marks the mean of 0.866; the dashed line marks a drop equal to the full dividend.

4.2 Is the Drop Smaller Than the Dividend?

Table 2: Tests of the Mean Drop Ratio Against the Full Dividend (1.0)

Measure	Mean	Clustered SE	95% CI	t vs 1.0	p-value	Naive t	Firm bootstrap 95% CI

Raw close-to-open	0.777	0.036	0.708 to 0.847	−6.25	4.0×10^{-10}	−11.84	0.735 to 0.817
Raw close-to-close	0.749	0.053	0.646 to 0.853	−4.76	2.0×10^{-6}	−8.51	0.685 to 0.811
Original adjustment, sign reversed	0.614	0.067	0.483 to 0.745	−5.77	7.8×10^{-9}	−13.52	0.549 to 0.678
Full-day market return, $\beta = 1$	0.915	0.044	0.829 to 1.001	−1.93	0.053	−4.02	0.876 to 0.956
Opening market return, $\beta = 1$	0.862	0.019	0.825 to 0.899	−7.27	3.6×10^{-13}	−9.42	0.829 to 0.897
Opening market return, estimated β (main)	0.866	0.018	0.831 to 0.901	−7.48	7.3×10^{-14}	−9.54	0.836 to 0.896
Close-to-close, full-day return, estimated β	0.930	0.032	0.867 to 0.993	−2.17	0.030	−2.79	0.876 to 0.985

Yes. On the main measure the average drop is 0.866 times the dividend, with a clustered 95% confidence interval of 0.831 to 0.901 that excludes 1.0 by a wide margin ($t = -7.48$, $p = 7.3 \times 10^{-14}$). The firm-cluster bootstrap gives nearly the same interval, and the Wilcoxon test rejects a median of 1.0 ($p = 7.8 \times 10^{-104}$). Prices fall by about 87 cents for every dollar of dividend.

Table 3: Robustness of the Market-Adjusted Shortfall

Check	Estimate	95% CI or SE	Test vs 1.0
Main measure (Table 2)	0.866	0.831 to 0.901	$t = -7.48$
No trim (all 19,319 events)	0.849	0.725 to 0.974	$t = -2.38$, $p = 0.017$
Excluding the 218 flat opens	0.869	0.833 to 0.904	$t = -7.29$

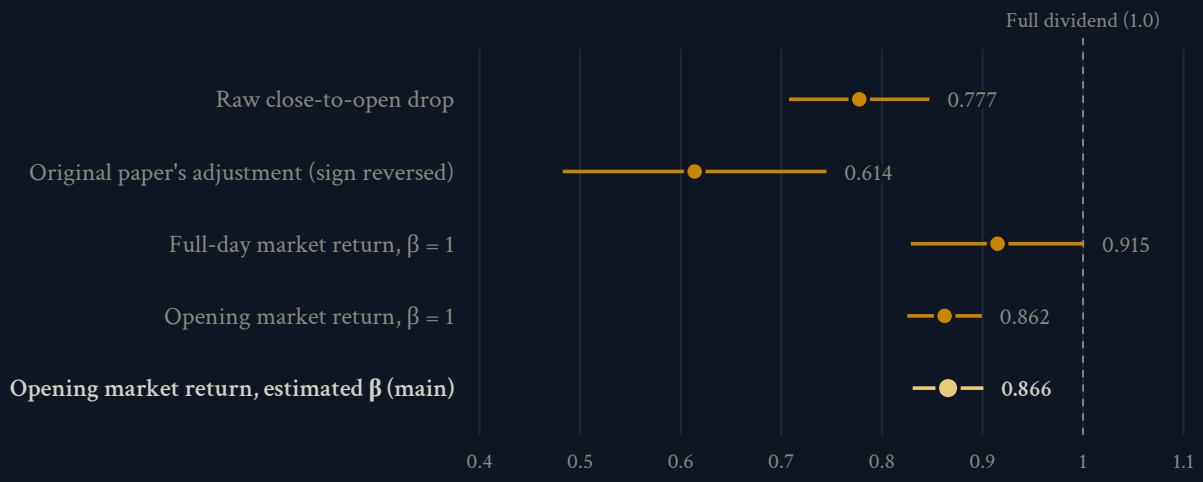
Median of the main measure	0.873		Wilcoxon $p = 7.8 \times 10^{-104}$
Dividend-weighted (pooled), market-adjusted	0.924	SE 0.027	$t = -2.85$, $p = 0.0044$
Dividend-weighted (pooled), raw	0.850	SE 0.023	

The shortfall does not depend on the trim, on the suspect flat opens, or on how events are weighted. Without any trim the mean is 0.849 ($p = 0.017$). Weighting events by dividend size, which estimates the dollars of price drop per dollar of dividend across the whole sample and is far less sensitive to tiny dividends, gives 0.924 (SE 0.027), still significantly below 1.0. The pooled figure is higher than the simple mean because large dividends belong to high-yield stocks, where the shortfall is smallest (Section 4.4).

4.3 Why the Market Adjustment's Timing Matters

Market returns on ex-dates were slightly positive on average: SPY's opening return averaged 0.024% and was positive on 55% of ex-dates, while its full-day return averaged 0.062%. Because the market term is multiplied by the price-to-dividend ratio (about 180 for a typical event), even these small returns move the ratio noticeably. Figure 2 shows how the answer depends on the adjustment.

FIGURE 2. THE MEAN DROP RATIO UNDER FIVE MEASURES



Dots are means; bars are 95% confidence intervals clustered by ex-date and firm. The main measure (bottom) uses the market's opening return and each stock's estimated beta.

Three points stand out. First, the original paper's adjustment, which subtracted the full-day market term, gives 0.614 on this sample, close to the 0.627 the original reported; this suggests the original computation and data were broadly reproducible, and that its error was the sign. Second, adding the full-day market return back with the correct sign, as the October 2026 revision did arithmetically, gives 0.915 with a median of 1.013, and its interval reaches 1.0 ($p = 0.053$). That version over-adjusts: it credits the stock with market gains made during the ex-day session, after the opening price was already set. Third, using only the market's overnight move, the return that can actually affect the open, gives 0.862 with $\beta = 1$ and 0.866 with estimated betas. The median beta in the sample is 0.93, so estimating betas changes little; the timing of the market return is what matters. The October revision's estimate of about 1.0 therefore came from the timing mismatch, and the real data reverse it.

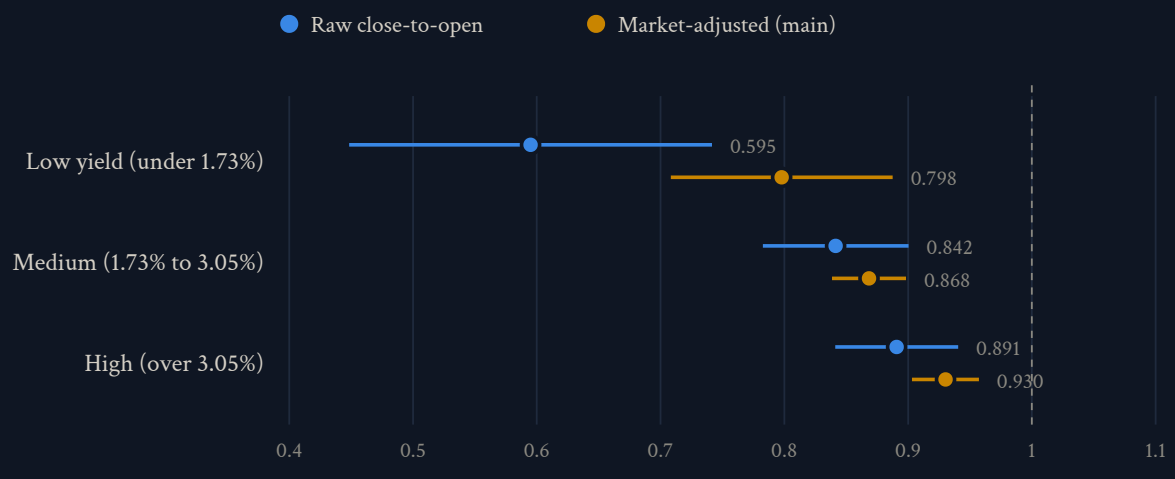
4.4 Dividend Yield

Table 4: Market-Adjusted Drop Ratio by Dividend Yield Tercile

Yield tercile	Trailing yield	N	Firms	Mean	Median	Clustered SE	t vs 1.0	p-value	Raw mean
Low	under 1.73%	6,348	302	0.798	0.795	0.046	−4.42	9.8×10^{-6}	0.595
Medium	1.73% to 3.05%	6,435	319	0.868	0.849	0.015	−8.62	6.9×10^{-18}	0.842
High	over 3.05%	6,435	257	0.930	0.919	0.014	−5.05	4.3×10^{-7}	0.891

The shortfall shrinks as yield rises: the adjusted ratio is 0.798 for the lowest-yield third of events, 0.868 for the middle third, and 0.930 for the highest, and the clustered Wald test rejects equal means ($\chi^2 = 15.0$ with 2 degrees of freedom, $p = 0.0006$). All three remain significantly below 1.0. The original paper found the same ordering in raw ratios but warned that it might be an artifact of the uncorrected market term, which weighs most on low-yield stocks. The adjustment does narrow the gap between the high and low terciles, from 0.296 in raw ratios to 0.132, but the gradient survives it, so it is not merely a by-product of the market's direction.

FIGURE 3. DROP RATIO BY YIELD TERCILE, RAW AND MARKET-ADJUSTED



Dots are means; bars are clustered 95% confidence intervals. The dashed line is the full dividend.

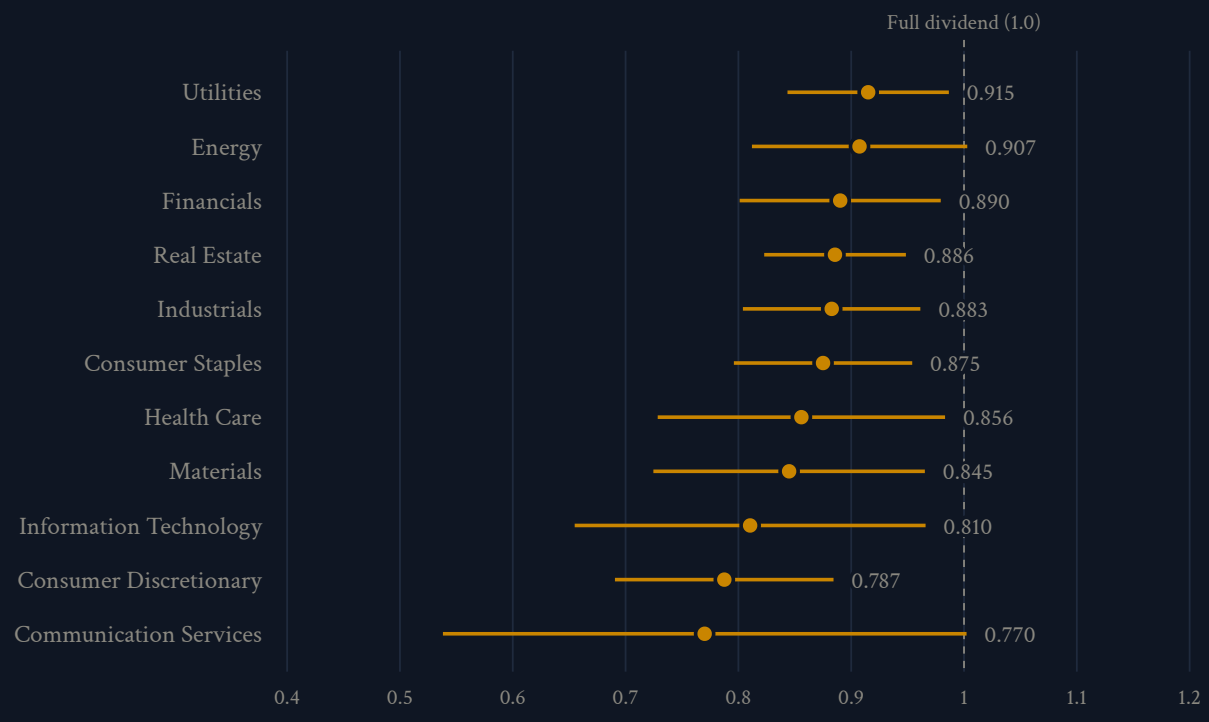
4.5 Sector

Table 5: Market-Adjusted Drop Ratio by Sector

Sector (GICS)	N	Firms	Mean	Median	Clustered SE	t vs 1.0	p-value
Utilities	1,627	31	0.915	0.929	0.037	−2.33	0.020
Energy	1,022	21	0.907	0.913	0.049	−1.90	0.057
Financials	3,393	70	0.890	0.866	0.045	−2.41	0.016
Real Estate	1,412	28	0.886	0.888	0.032	−3.57	0.0004
Industrials	3,315	71	0.883	0.850	0.040	−2.92	0.0035
Consumer Staples	1,641	29	0.875	0.874	0.040	−3.10	0.0019
Health Care	1,879	39	0.856	0.850	0.065	−2.22	0.027
Materials	1,157	23	0.845	0.868	0.061	−2.53	0.012
Information Technology	1,777	40	0.810	0.893	0.079	−2.39	0.017
Consumer Discretionary	1,594	34	0.787	0.808	0.049	−4.30	1.7×10^{-5}
Communication Services	401	12	0.770	0.867	0.118	−1.94	0.052

Every sector's mean is below 1.0, ranging from 0.770 in Communication Services to 0.915 in Utilities, and 9 of the 11 are significantly below it at the 5% level. The differences between sectors, however, are not statistically significant (Wald $\chi^2 = 7.00$ with 10 degrees of freedom, $p = 0.725$), so sector membership adds little once yield is known. Sectors here are each company's current GICS classification.

FIGURE 4. MARKET-ADJUSTED DROP RATIO BY SECTOR



Sorted from highest to lowest mean. Bars are clustered 95% confidence intervals.

4.6 Time Series

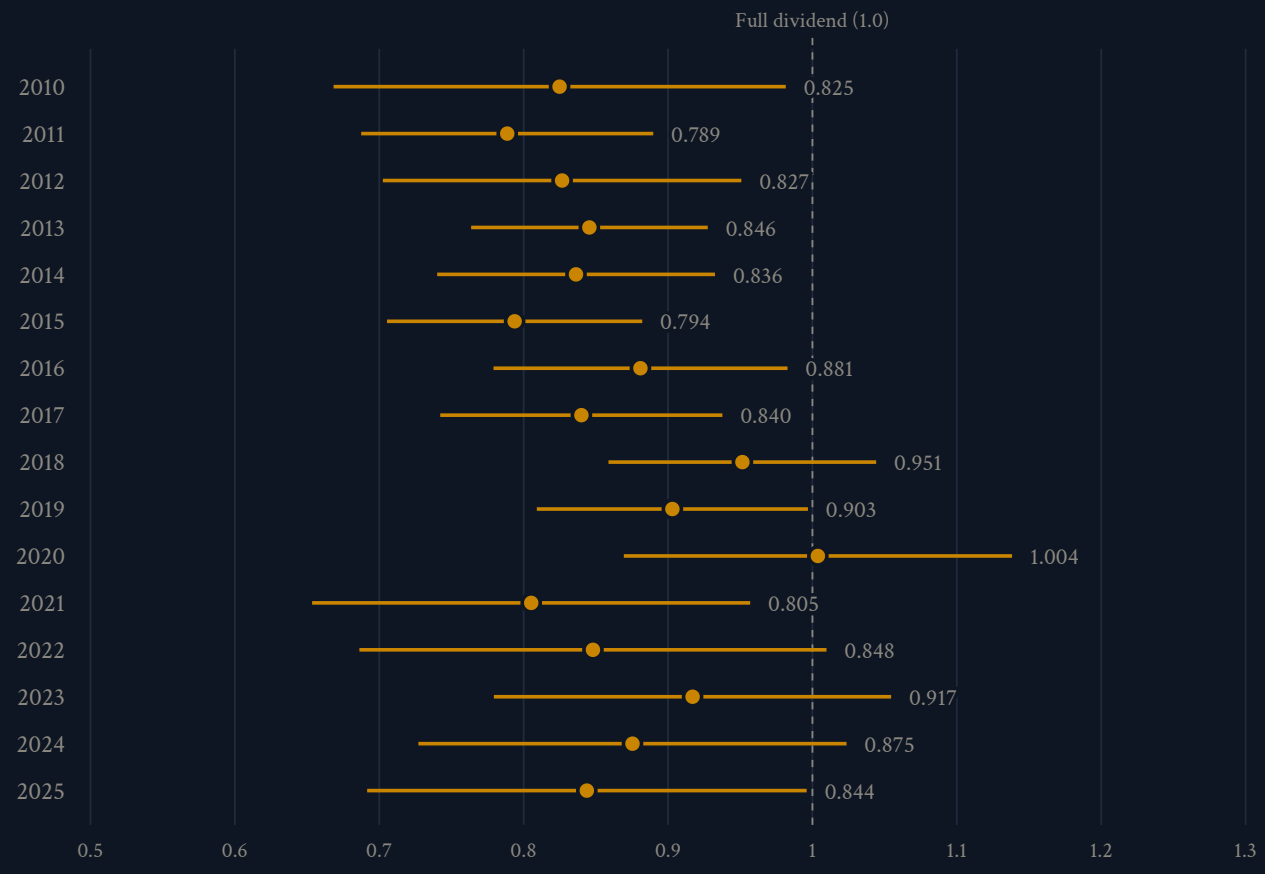
Table 6: Market-Adjusted Drop Ratio by Year

Year	N	Firms	Mean	Median	Clustered SE	t vs 1.0	p-value	Raw mean
2010	840	222	0.825	0.763	0.080	−2.19	0.028	0.520
2011	868	237	0.789	0.807	0.052	−4.10	4.2×10^{-5}	0.841
2012	956	252	0.827	0.742	0.063	−2.74	0.0062	0.766
2013	1,000	262	0.846	0.851	0.042	−3.69	0.0002	0.566
2014	1,058	272	0.836	0.863	0.049	−3.33	0.0009	0.903
2015	1,109	284	0.794	0.846	0.045	−4.58	4.8×10^{-6}	0.807

2016	1,164	300	0.881	0.883	0.052	−2.29	0.022	0.906
2017	1,230	316	0.840	0.861	0.050	−3.21	0.0013	0.716
2018	1,262	324	0.951	0.874	0.047	−1.03	0.305	0.750
2019	1,321	339	0.903	0.886	0.048	−2.03	0.043	0.689
2020	1,286	344	1.004	1.053	0.069	0.06	0.956	1.096
2021	1,316	338	0.805	0.882	0.077	−2.52	0.012	0.551
2022	1,384	354	0.848	0.898	0.083	−1.84	0.065	1.056
2023	1,438	367	0.917	0.915	0.070	−1.18	0.237	0.756
2024	1,469	381	0.875	0.921	0.076	−1.65	0.100	0.476
2025	1,517	390	0.844	0.937	0.078	−2.01	0.044	0.973

The adjusted ratio is below 1.0 in 15 of 16 years and significantly so in 11; the exceptions, where the 95% interval includes 1.0, are 2018, 2020, 2022, 2023, and 2024. The year-to-year differences are not significant (Wald p = 0.320), and there is no trend. The raw ratios swing much more, from 0.476 to 1.096, because they move with the market's direction on each year's ex-dates; the adjustment removes most of that swing.

FIGURE 5. MARKET-ADJUSTED DROP RATIO BY YEAR



Bars are clustered 95% confidence intervals.

4.7 Yield and Share Price Regressions

Table 7: Regressions and Correlations

Relationship	Estimate	Clustered SE	p-value
Main measure on trailing yield (per percentage point)	0.034	0.015	0.022
Raw close-to-open on trailing yield (per percentage point)	0.065	0.017	0.0002
Main measure on log nominal share price	−0.016	0.027	0.559
Spearman correlation, main measure and yield	0.052		5.3×10^{-13}

Spearman correlation, main measure and nominal price	−0.011	0.135

Each percentage point of trailing yield adds about 0.034 to the adjusted ratio (p = 0.022), half the slope in raw ratios. The relation is real but explains very little of the event-to-event variation. Nominal share price, measured at the prices that actually traded rather than split-adjusted ones, shows no relationship with the drop ratio: by price bucket the means are under $25 0.824, $25 to $50 0.912, $50 to $100 0.868, $100 to $200 0.847, over $200 0.841. A tick-size explanation would predict a larger shortfall for low-priced stocks, which the data do not show.

4.8 Which Stocks Show It Most

Because each stock contributes only about 60 events, a single stock's average is far noisier than the sample average, and ranking stocks by their raw means mostly ranks noise: the extremes in either direction are stocks with very small dividends relative to their price, such as Nvidia. We therefore test each stock with at least 20 events against 1.0 and control the false discovery rate across all of them.

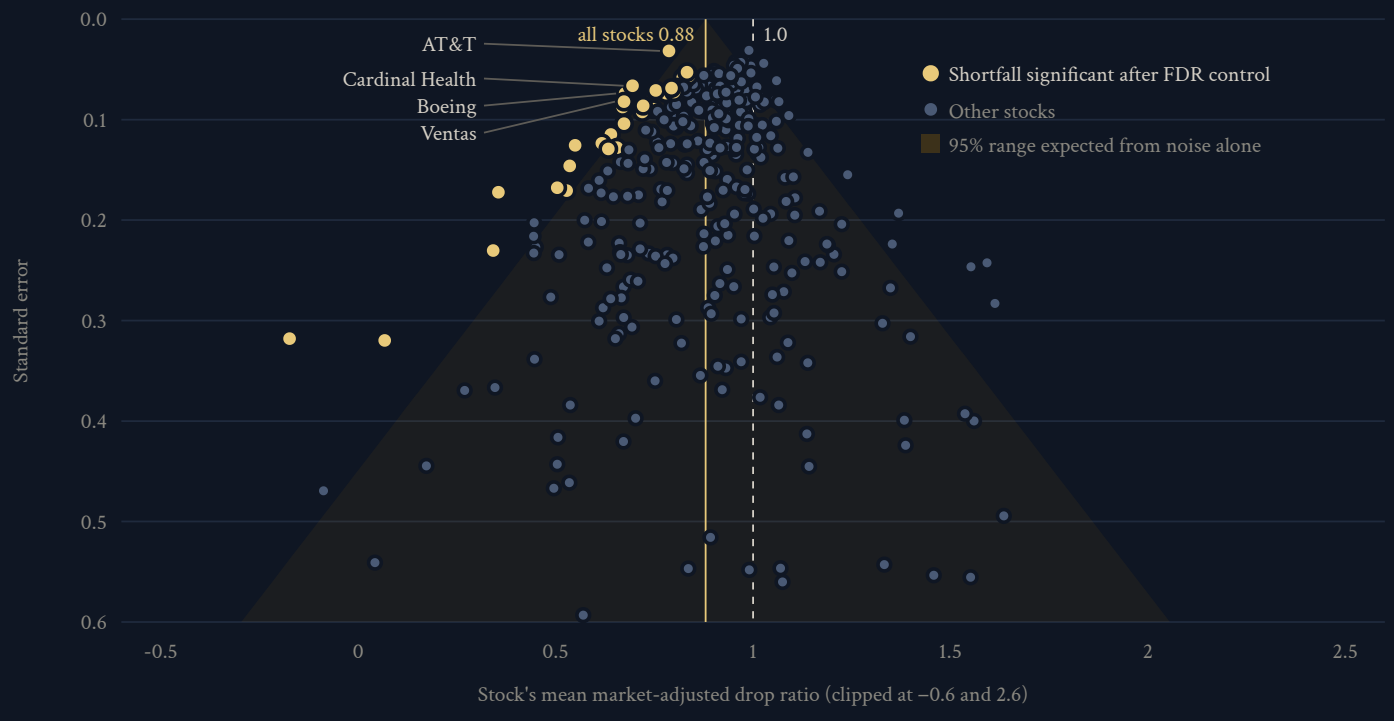
Table 9: Stock-Level Results

Statistic	Value
Stocks with at least 20 events	343 (median 62 events each)
Mean of stock-level means	0.874
Significantly below 1.0 at p < 0.05	65 stocks (about 9 expected by chance)
Significantly above 1.0 at p < 0.05	4 stocks (about 9 expected by chance)
Below 1.0 after FDR control (q < 0.10)	32 stocks
Above 1.0 after FDR control (q < 0.10)	0 stocks
Typical standard error of a stock's mean	0.150

Estimated spread of true stock-level ratios (SD)	0.155
Odd-year vs even-year stock means, correlation	0.17 (Spearman 0.09)

The shortfall shows up far more often than chance allows. 65 of 343 stocks are significantly below 1.0 at the 5% level, against roughly 9 that would be expected if every stock's true ratio were 1.0, while only 4 are significantly above it. After false discovery rate control, 32 stocks retain a significant shortfall and none a significant excess. The firm-level differences are jointly significant, but they are modest: the empirical Bayes model puts the spread of true stock-level ratios at about 0.15 around a center of 0.88, about the same size as the noise in a typical stock's mean (0.15). Consistent with that, a stock's average in odd years predicts its average in even years only weakly (correlation 0.17).

FIGURE 6. STOCK-LEVEL MEANS AGAINST THEIR PRECISION



Each dot is one of the 343 stocks with at least 20 events; stocks lower on the chart have noisier means. The shaded band is the 95% range a stock's mean would fall in from noise alone if every stock shared the sample center of 0.88. Gold dots are stocks with a significant shortfall after false discovery rate control.

Table 8: The 32 Stocks With a Significant Shortfall After False Discovery Rate Control (q < 0.10)

Ticker	Company	Sector	Events	Mean	Pooled	t vs 1.0	FDR q
T	AT&T	Communication Services	64	0.787	0.763	−6.75	0.000
CAH	Cardinal Health	Health Care	63	0.695	0.696	−4.60	0.004
BA	Boeing	Industrials	41	0.676	0.652	−4.38	0.010
VTR	Ventas	Real Estate	66	0.673	0.749	−3.98	0.015
PKG	Packaging Corporation of America	Materials	34	0.670	0.689	−3.78	0.033
EBAY	eBay Inc.	Consumer Discretionary	28	0.355	0.403	−3.74	0.035
HUM	Humana	Health Care	53	-0.174	-0.298	−3.69	0.033
TT	Trane Technologies	Industrials	60	0.549	0.381	−3.59	0.033
WM	Waste Management	Industrials	64	0.753	0.729	−3.48	0.035
PSA	Public Storage	Real Estate	64	0.721	0.732	−3.23	0.066
EQIX	Equinix	Real Estate	43	0.535	0.532	−3.18	0.066
PEP	PepsiCo	Consumer Staples	64	0.833	0.835	−3.18	0.066
SYK	Stryker Corporation	Health Care	64	0.673	0.616	−3.14	0.066
CSX	CSX Corporation	Industrials	64	0.640	0.582	−3.14	0.066
KR	Kroger	Consumer Staples	64	0.617	0.491	−3.10	0.066

MCD	McDonald's	Consumer Discretionary	64	0.822	0.840	−3.06	0.070
DOV	Dover Corporation	Industrials	64	0.719	0.701	−3.04	0.070
XOM	ExxonMobil	Energy	64	0.793	0.800	−3.01	0.071
HON	Honeywell Technologies	Industrials	64	0.780	0.734	−3.00	0.071
RF	Regions Financial Corporation	Financials	60	0.504	0.730	−2.96	0.077
CBOE	Cboe Global Markets	Financials	34	0.067	0.020	−2.92	0.085
MRK	Merck & Co.	Health Care	64	0.837	0.827	−2.88	0.085
IVZ	Invesco	Financials	64	0.827	0.862	−2.86	0.085
GM	General Motors	Consumer Discretionary	39	0.342	0.597	−2.86	0.088
SNA	Snap-on	Industrials	64	0.633	0.677	−2.84	0.085
NEE	NextEra Energy	Utilities	64	0.807	0.802	−2.83	0.085
BNY	BNY Mellon	Financials	64	0.747	0.780	−2.82	0.085
NVDA	Nvidia	Information Technology	41	-1.827	-2.477	−2.78	0.095
ADI	Analog Devices	Information Technology	64	0.528	0.420	−2.77	0.091
WY	Weyerhaeuser	Real Estate	60	0.800	0.818	−2.76	0.091
BXP	BXP, Inc.	Real Estate	61	0.837	0.830	−2.70	0.097
NUE	Nucor	Materials	64	0.655	0.618	−2.69	0.097

The most consistent shortfalls belong to large, steady quarterly payers. AT&T, the clearest case, averaged 0.787 over 64 events (t = −6.75); Cardinal Health, Boeing, Ventas, and Packaging Corporation of America follow, and the list also includes PepsiCo, McDonald's, Merck, ExxonMobil, Honeywell, and NextEra Energy. The 32 names span all 11 sectors (Industrials 7, Real Estate 5, Health Care 4, Financials 4, Consumer Discretionary 3, Materials 2, Consumer Staples 2, Information Technology 2, Communication Services 1, Energy 1, Utilities 1) and have a median trailing yield of 2.5%, close to the 2.3% of the full stock set, so they are not simply the low-yield names. The pooled column, which divides each stock's total adjusted price drop by its total dividends, agrees with the simple mean for nearly all of them. Given the weak year-to-year persistence, the list identifies where the shortfall has been most reliable over 2010 to 2025, not a ranking expected to repeat.

4.9 A Direct Test of Dividend Capture

The original paper did not test a capture trade. We do so directly: buy at the close before the ex-date, collect the dividend, and sell at the ex-day open, with the return measured against the purchase price.

Table 10: Overnight Dividend Capture, Per Event, Before Costs and Taxes (N = 19,319)

Version	Mean (bps)	Clustered SE (bps)	t vs 0	Median (bps)	Share positive
Gross	9.03	1.75	5.15	10.87	57.8%
Gross, net of the stock's expected market move	6.32	0.68	9.26	7.12	58.1%

The overnight trade earned 9.03 basis points per event on average, or 6.32 basis points after removing the stock's expected market move, and was profitable 58% of the time. That margin is the dividend shortfall seen from the other side and is precisely estimated, but it is small: it must cover the bid-ask spread paid on entry and exit, any commissions, and, for a taxable U.S. investor, ordinary income tax

on a dividend held too briefly to qualify for the reduced rate (qualified treatment requires holding the shares more than 60 days during the 121-day period that begins 60 days before the ex-date). None of these costs is modeled here, so the result does not show that the trade is profitable after costs.

SECTION V

5. Discussion

5.1 Taxes

Under the Elton and Gruber (1970) framework, a ratio of 0.866 would mean the marginal investor values a dollar of dividends at about 87 cents relative to a dollar of capital gains:

$$(1 - \tau_d) / (1 - \tau_g) \approx 0.87$$

For most U.S. individuals, current law makes that hard to produce. Qualified dividends and long-term capital gains are taxed at the same reduced rates, short-term traders pay ordinary rates on both, and tax-exempt and tax-deferred accounts pay neither, so each of these groups should be close to indifferent and push the ratio toward 1.0. Investors who do prefer gains to dividends include foreign holders, whose U.S. dividends are subject to withholding while their U.S. capital gains generally are not; at a 15% treaty withholding rate, such an investor values a dividend at about 85 cents on the dollar, close to the estimate here. Taxes therefore remain a candidate explanation, but only through investors outside the domestic individual clientele the original paper had in mind. The yield gradient fits a clientele story, since income-oriented and tax-advantaged holders concentrate in high-yield stocks, but the same pattern also follows from the microstructure explanation below.

5.2 Market Microstructure

Frank and Jagannathan (1998) showed that bid-ask bounce alone can produce a shortfall: if cum-day trades cluster at the bid and ex-day trades at the ask, the measured drop is reduced by up to a full spread. Because the spread is roughly fixed in cents while dividends vary, that effect is proportionally largest for small dividends, which matches the yield gradient in Table 4. Graham, Michaely, and Roberts (2003), on the other hand, found the drop ratio fell as tick sizes shrank, which argues against microstructure as the main force. Our test on nominal prices (Table 7) finds no sign of a tick-size effect. Separating the bid-ask explanation from the tax explanation would require quote data, which this study does not use.

5.3 Comparison With Prior Literature

Study	Period	Market	Drop ratio
Elton & Gruber (1970)	1966 to 1967	US	about 0.78
Frank & Jagannathan (1998)	1980 to 1993	Hong Kong	about 0.50
Graham et al. (2003)	2001	US (decimal era)	0.750 (median, close-to-close)
This study, raw	2010 to 2025	US	0.777 (mean, close-to-open); 0.810 (median, close-to-close)
This study, market-adjusted	2010 to 2025	US	0.866 (mean, close-to-open)

The raw ratios sit in the range of the earlier U.S. literature. The comparison is loose, because the earlier figures are mostly unadjusted close-to-close measures; this study's own close-to-close median, 0.810, is the closest like-for-like figure to Graham et al.'s 0.750.

5.4 Limitations

Several limitations remain. The sample is point-in-time on entry but not on exit: firms that left the S&P 500 before November 2026 are missing. Those firms include acquired, distressed, and demoted

companies, whose ex-day behavior may differ, so the estimates describe companies that remained in the index; a database with delisted securities, such as CRSP, would close this gap. Yahoo's opening prints are not audited; the flat-open check in Table 3 suggests the affected events do not change the result. SPY's opening print reflects the market at 9:30 a.m., while a stock's own opening trade can come a little later, and betas are estimated from close-to-close returns but applied to an overnight return. The ±20 trim removes 101 events from the main measure, and the untrimmed result points the same way. Sector labels are current GICS classifications. Finally, the capture test ignores trading costs and taxes, and no test here uses quote data, so the competing explanations in Sections 5.1 and 5.2 cannot be told apart.

SECTION VI

6. Implications

6.1 For Estimating the Ex-Day Price

On average, a stock in this sample opened 0.866 times its dividend below the prior close, after allowing for the market's overnight move, and the figure ran from about 0.80 for low-yield stocks to 0.93 for high-yield stocks. For any single event the noise is far larger than the effect, so these are averages, not forecasts of a given stock's open.

6.2 For Dividend Capture

The shortfall leaves a small, statistically reliable overnight margin of about 6.32 basis points per event net of the market. Spreads, commissions, and the ordinary-income treatment of briefly held dividends all come out of that margin, and this study does not measure them.

6.3 For Market Efficiency

A drop of about 87% of the dividend is a real departure from the frictionless benchmark, but it is the kind of gap that frictions are expected to leave: it is small in dollars, and the trade that would close it carries the costs described above. The results are consistent with an efficient market with frictions and are not evidence of a free arbitrage.

SECTION VII

7. Conclusions

Using 19,319 ex-dividend events from 399 S&P 500 firms over 2010 to 2025, counted only while each firm was in the index, we find that share prices fall by less than the full dividend. Before adjustment the open is lower by 0.777 times the dividend; after adding back each stock's expected overnight market move, the drop is 0.866 times the dividend (95% CI 0.831 to 0.901, clustered by date and firm). The result holds without trimming, without suspect opening prints, and when events are weighted by dividend size.

The shortfall is:

1. Larger for low-yield stocks (0.798) than for high-yield stocks (0.930), a gradient that survives the market adjustment ($p = 0.0006$).
2. Present in every sector, with no significant differences between them ($p = 0.725$).
3. Present in 15 of 16 years and significant in 11, with no trend ($p = 0.320$ for differences across years).
4. Unrelated to nominal share price.
5. Concentrated, at the stock level, in a group of 32 stocks with significant shortfalls after false discovery rate control, led by AT&T, Cardinal Health, Boeing, and Ventas, with no stock showing a significant excess.

The October 2026 revision of this paper concluded, from the original's own figures, that the shortfall disappeared after a correctly signed market adjustment. Run on the real prices, that conclusion does not hold: the October figure added back the market's full-day return, which includes moves after the open, and using the overnight move instead leaves a clear shortfall of about 13% of the dividend. Foreign investors' dividend withholding and bid-ask bounce are both consistent with its size and its yield gradient; telling them apart would require ownership or quote data.

Revision note, November 2026. This version reruns the study on real Yahoo Finance prices. The original 80-stock sample was never listed, so it is replaced by a point-in-time S&P 500 sample of 19,319 events, with the ticker list published below. The market adjustment now uses SPY's opening return and estimated betas; yields are trailing twelve-month yields; standard errors are clustered by date and firm; the trim's effect is reported; the price-level test uses nominal prices; and stock-level results and a direct capture test are added. The October 2026 revision had corrected the sign of the original adjustment and estimated a corrected mean of about 1.0 from the original's reported figures; the rerun shows that estimate was an artifact of using the full-day market return, and the corrected mean is 0.866. All tables, figures, and conclusions in Sections 3 to 7 are new; Sections 1 and 2 are unchanged apart from minor edits.

APPENDIX

Data Appendix

The 399 tickers in the final sample, as listed in the S&P 500 on November 5, 2026 (EQR appears as VMRK following its 2026 merger with AvalonBay):

A, AAPL, ABBV, ABT, ACGL, ACN, ADI, ADM, ADP, AEE, AEP, AES, AFL, AIG, AIZ, AJG, ALB, ALL, ALLE, AMAT, AMCR, AME, AMGN, AMP, AMT, AON, AOS, APA, APD, APH, APO, APTV, ARE, ARES, ATO, AVGO, AVY, AWK, AXP, BA, BAC, BALL, BAX, BBY, BDX, BEN, BF-B, BG, BKNG, BKR, BLK, BMY, BNY, BR, BRO, BX, BXP, C, CAH, CARR, CAT, CB, CBOE, CCI, CCL, CDW, CEG, CF, CFG, CHD, CHRW, CI, CINF, CL, CLX, CMCSA, CME, CMI, CMS, CNP, COF, COO, COP, COR, COST, CPT, CRM, CSCO, CSX, CTAS, CTSH, CVS, CVX, D, DAL, DD, DE, DELL, DG, DGX, DHI, DHR, DIS, DLR, DOC, DOV, DOW, DPZ, DRI, DTE, DUK, DVN, EBAY, ECL, ED, EFX, EG, EIX, EL, ELV, EME, EMR, EOG, EQIX, EQT, ERIE, ES, ESS, ETN, ETR, EVRG, EXC, EXE, EXPD, EXPE, EXR, F, FANG, FAST, FCX, FDS, FDX, FE, FIS, FITB, FOXA, FRT, FTV, GD, GE, GEHC, GEN, GEV, GILD, GIS, GL, GLW, GM, GOOGL, GPC, GPN, GRMN, GS, GWW, HAL, HAS, HBAN, HCA, HD, HIG, HII, HLT, HON, HPE, HPQ, HRL, HST, HSY, HUBB, HUM, HWM, IBKR, IBM, ICE, IEX, IFF, INTC, INTU, INVH, IP, IR, IRM, ITW, IVZ, J, JBHT, JBL, JCI, JKHY, JNJ, JPM, KDP, KEY, KHC, KIM, KKR, KLAC, KMB, KMI, KO, KR, L, LDOS, LEN, LH, LHX, LII, LIN, LLY, LMT, LNT, LOW, LRCX, LUV, LVS, LYB, MA, MAA, MAR, MAS, MCD, MCHP, MCK,

MCO, MDLZ, MDT, MET, META, MGM, MKC, MLM, MMM, MO, MOS, MPC, MPWR, MRK, MRSH, MS, MSCI, MSFT, MSI, MTB, MU, NDAQ, NDSN, NEE, NEM, NI, NKE, NOC, NRG, NSC, NTAP, NTRS, NUE, NVDA, NWSA, NXPI, O, ODFL, OKE, OMC, ORCL, OTIS, OXY, PAYX, PCAR, PCG, PEG, PEP, PFE, PFG, PG, PGR, PH, PHM, PKG, PLD, PM, PNC, PNR, PNW, PPG, PPL, PRU, PSA, PSX, PWR, PYPL, Q, QCOM, RCL, REG, REGN, RF, RJF, RL, RMD, ROK, ROL, ROP, ROST, RSG, RTX, RVTY, SBAC, SBUX, SCHW, SHW, SJM, SKYD, SLB, SNA, SO, SPG, SPGI, SRE, STE, STLD, STT, STX, STZ, SW, SWK, SWKS, SYF, SYK, SYY, T, TDG, TECH, TEL, TER, TFC, TGT, TJX, TKO, TMO, TMUS, TPL, TPR, TRGP, TROW, TRV, TSCO, TSN, TT, TXN, TXT, UDR, UHS, UNH, UNP, UPS, URI, USB, V, VICI, VLO, VLTO, VMC, VMRK, VRSK, VRSN, VST, VTR, VTRS, VZ, WAB, WDC, WEC, WELL, WFC, WM, WMB, WMT, WRB, WSM, WST, WTW, WY, WYNN, XEL, XOM, XYL, YUM, ZBH, ZTS

Prices, dividends, and splits: Yahoo Finance chart service, daily data, retrieved November 10, 2026. Index membership and addition dates: datasets/s-and-p-500-companies (GitHub), snapshot of November 5, 2026. Analysis in Python with pandas, NumPy, SciPy, and statsmodels. Data and analysis code are available from the author on request.

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